@duck

TCM SOC 101 SIEM

- SIEM
- Architecture
- Deployment Models
- Logs Types and Formats
- Common Attack Behaviour
- Log analysis
- Splunk

What is SIEM

Architecture

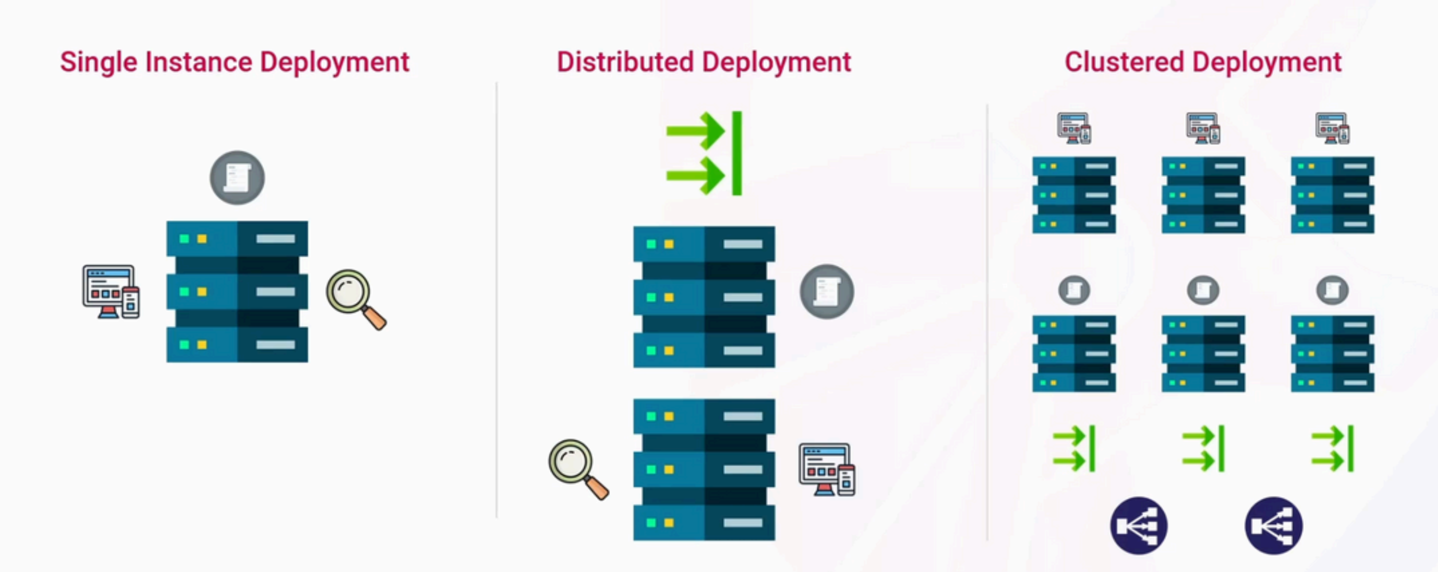
1. Log Managment
2. Real-time Monitoring
3. Alerting and Notification
4. Incident Response
5. Dashboard, Reports and Visualization
6. Threat Intelligence Integration

Deployment Models

Single Instance Deployment

Distributed Deployment

Clustered Deployment



LOG TYPES

Log Types

- System Logs
  - Windows Event Logs
  - Sysmon Logs
  - Linux/Unix Syslogs
- Network Logs
  - Firewall Logs
  - Proxy Logs
  - DNS Logs
- Application Logs
  - Database Logs
  - Web Server / HTTP Logs
  - App Logs
- Security Logs
  - Authentication Logs
  - IDS/IPS Logs
  - Endpoint Security Logs

Log Formats

- **Unstructured Logs**
  - No predefined format or syntax
  - Common Log Format (CLF)
- **Semi-Structured Logs**
  - Some syntax structure
  - Lack adherence to a schema
  - Syslog
  - Windows Event Log (EVT)
- **Structured Logs**
  - Well-defined syntax and formatting
  - Adherence to an agreed upon schema
  - CSV, TSV
  - JavaScript Object Notation (JSON)
  - eXtensible Markup Language (XML)


```
- <System>
  <Provider Name="Microsoft-Windows-Security-Auditing" Guid="{57249625-5478-4994-
a5ba-3e3b0328c30d}" />
  <EventID>4624</EventID>
  <Version>3</Version>
  <Level>0</Level>
  <Task>12544</Task>
  <Opcode>0</Opcode>
  <Keywords>0x8020000000000000</Keywords>
  <TimeCreated SystemTime="2024-07-12T01:38:40.1074714Z" />
  <EventRecordID>1133328</EventRecordID>
  <Correlation ActivityID="{a2179f13-bd97-000e-3d9f-17a297bdda01}" />
  <Execution ProcessID="1212" ThreadID="9936" />
  <Data Name="SubjectUserSid">S-1-5-18</Data>
  <Data Name="SubjectUserName">WKSTN-10293$</Data>
  <Data Name="SubjectDomainName">WORKGROUP</Data>
  <Data Name="SubjectLogonId">0x3e7</Data>
  <Data Name="TargetUserSid">S-1-5-18</Data>
  <Data Name="TargetUserName">SYSTEM</Data>
  <Data Name="TargetDomainName">NT AUTHORITY</Data>
  <Data Name="TargetLogonId">0x3e7</Data>
  <Data Name="LogonType">5</Data>
  <Data Name="LogonProcessName">Advapi</Data>
  <Data Name="AuthenticationPackageName">Negotiate</Data>
  <Data Name="WorkstationName">-</Data>
  <Data Name="ProcessName">C:\Windows\System32\services.exe</Data>
  <Data Name="IpAddress">-</Data>
  <Data Name="ImpersonationLevel">%%1833</Data>
  </EventData>
  </Event>
```


COMMON ATTACK Signatures

User Behavior Indicators

- **Multiple Failed Login Attempts**
  - Incorrect usernames or passwords
  - Increase in failures from a single user account
  - Increase in failures from multiple user accounts
- **Login Times**
  - Time of day that logons or access requests are taking place
  - Abnormalities from a user's baseline
- **Login/Access Locations**
  - Geographic locations of logons or access requests
  - Unusual countries or regions
  - Impossible Travel
- **File Access Patterns**
  - File paths, modifications, or other activity
- **User-Agent Strings**
  - Unusual or associated with known tools

SQL Injection

SQL Injection

- Inserting or injecting malicious SQL statements
- Manipulate expected database queries
  - Retrieve sensitive information
  - Extract hashed passwords
  - Bypass authentication logic
- Look for SQL keywords
  - `SELECT`, `FROM`, `WHERE`, `UNION`
- Look for injection characters
  - Single/double quotes, semicolons, comment indicators (dashes)
- URL-encoded keywords/injection characters
- Malformed entries or errors within database logs

- `SELECT`
  - Retrieve data from a database table
- `FROM`
  - Specifies the table to retrieve data from
- `WHERE`
  - Filter records based on conditions
- `ORDER BY`
  - Sort the results (ascending or descending)
- `GROUP BY`
  - Group rows with identical values into summaries

Data Manipulation

- `INSERT INTO`
  - Adds new rows of data into a table
- `UPDATE`
  - Modifies existing data in a table
- `DELETE`
  - Removes existing rows from a table
- `DROP`
  - Deletes a table or database from the system
- `JOIN`
  - Combines rows from two or more tables based on a colu
- `INNER JOIN`
  - Returns records that have matching values in multiple ta
- `UNION`
  - Combines the results of two or more SELECT statements

Comparative Operations

- `LIKE`
  - Searches for a specified pattern in a column
- `IN`
  - Checks if a value matches any value in a list or query

Cross Site Scripting

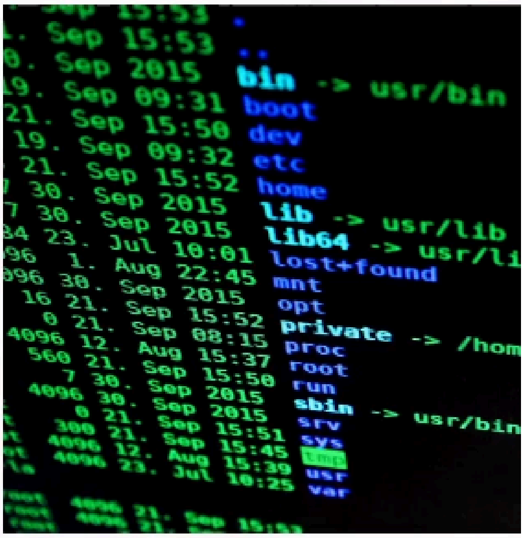
Cross-Site Scripting

- Executing malicious code by injecting JavaScript
  - Hijack user sessions
  - Steal cookies
  - Deface websites
- Look for <script> tag indicators
- Look for event handlers
  - onload, onclick, onmouseover
  - References to <script> tags
  - References to "javascript"
- Special characters
  - <, >, ", ', &, %, ;
- URL-encoded injection characters

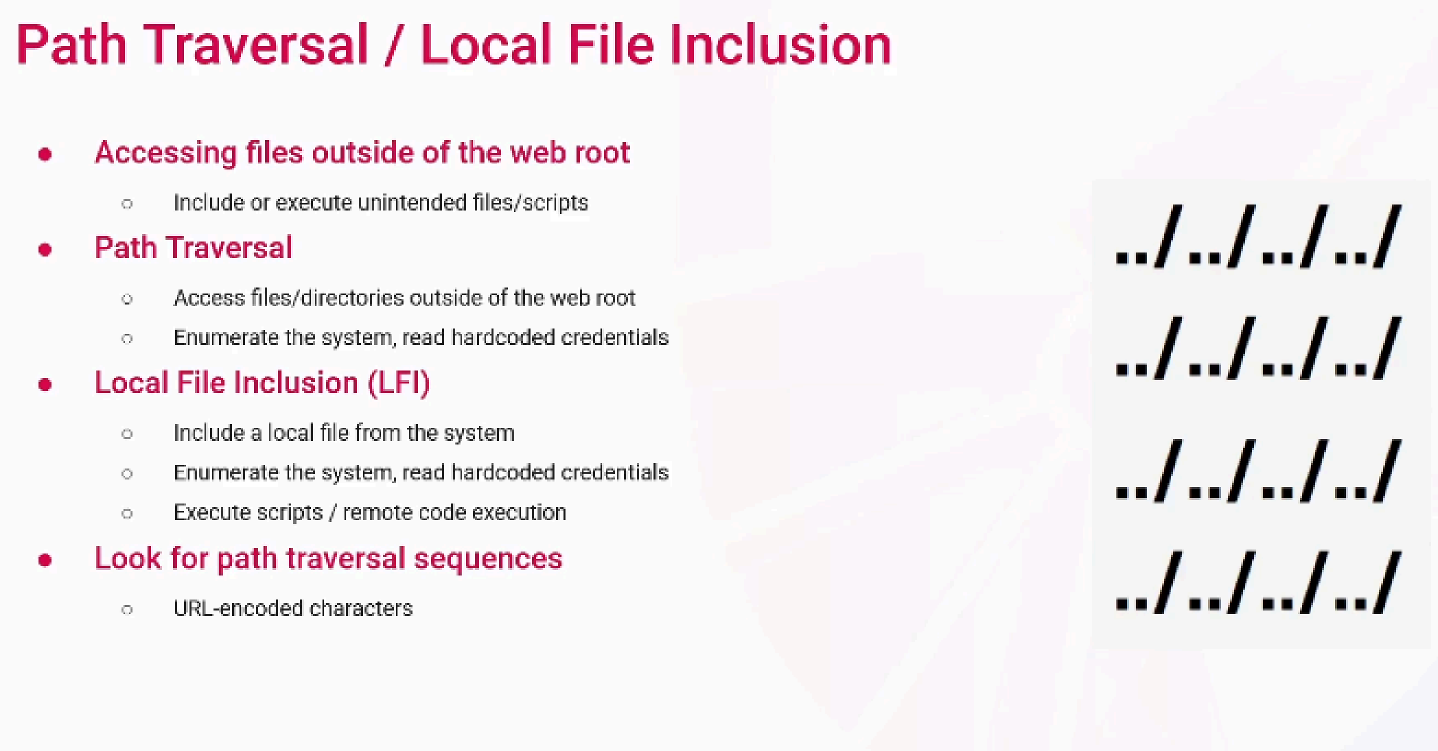
Command Injection

Command Injection

- Executing arbitrary OS commands
- Look for special characters that separate commands
  - ; , || , &&



Path traversal and local file inclusion



COMMAND LINE LOG ANALYSIS

1. file log.log
2. ls -lh log.log
3. wc -l log.log
4. wc -w log.log
5. wc -c log.log
6. head log.log -n 1
7. tail log.log -n 1
8. cut log.log -d “ ” -f 1
9. cut log.log -d “ ” -f 1 | sort | uniq -c
10. cut log.log -d “ ” -f 1 | sort | uniq -c | sort -nr
11. cut log.log -d “ ” -f 1 | sort | uniq -c | grep -v “ 1 ” | sort -nr

PATTERN MATCHING

1. grep “404” log.log
2. grep -c “404” log.log
3. grep -n “404” log.log
4. grep -E ‘%3C | %3E | < | >’ log.log

STRUCTURE LOG ANALYSIS

1. file event.json
2. jq . event.json

3. jq ‘length’ event.json
4. jq ‘map(.event)’ event.json
5. jq ‘.[] | select(.event.PROCESS_ID == 000)’ event.json
6. jq ‘.[] | select(.event.PROCESS_ID == 000)| .event.FILE_PATH’ event.json
7. jq ‘.[] | select(.event.PROCESS_ID == 000)| .event.HASH’ event.json
8. jq ‘.[] | select(.event.PROCESS_ID == 000)| .event.PROCESS_ID’ event.json

SPLUNK

Managing Splunk on Your Lab Machine

Now that we□re finished with the SIEM section, if you installed Splunk on your regular lab machine, feel free to stop the service and disable it from auto-starting to save on resources.

You can always re-enable it if you want to start it up again and get more practice.

To stop the Splunk service, use:

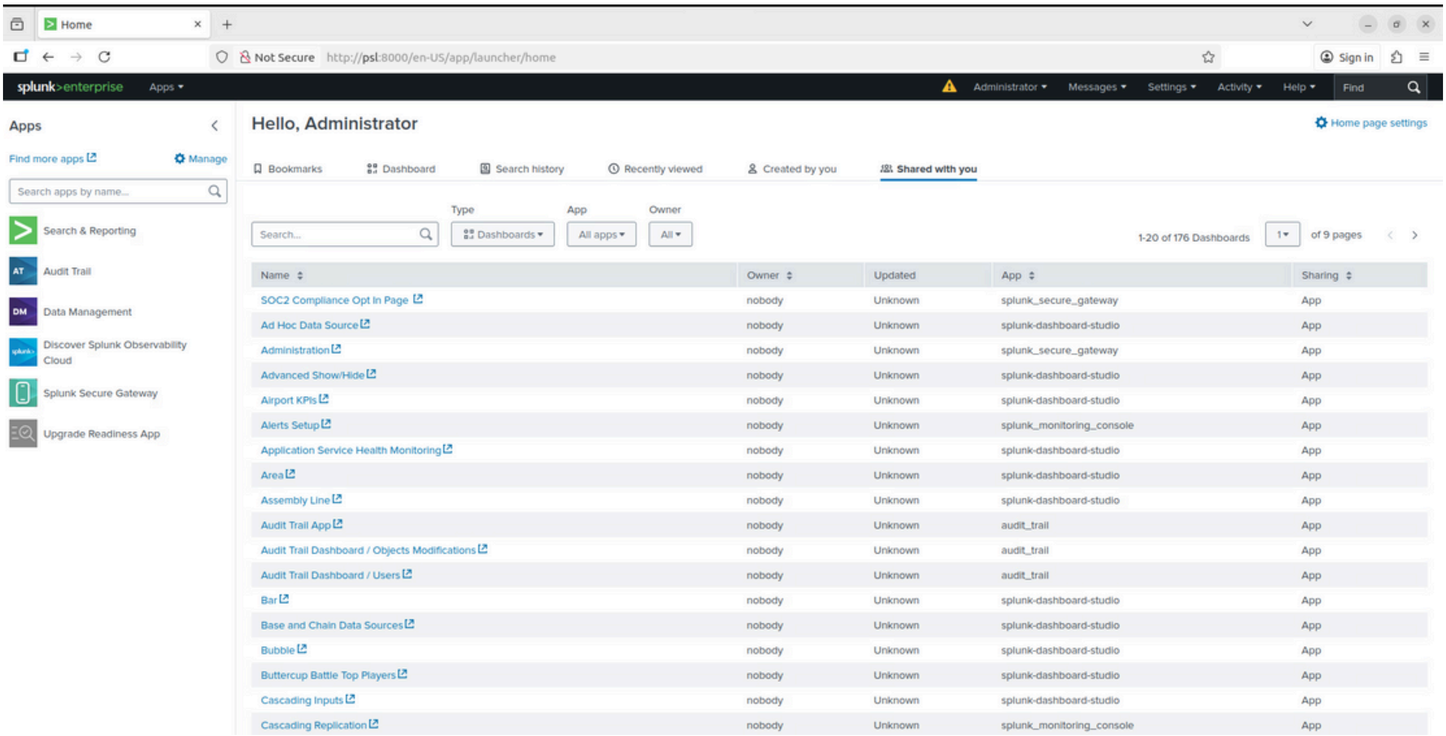
sudo /opt/splunk/bin/splunk stop

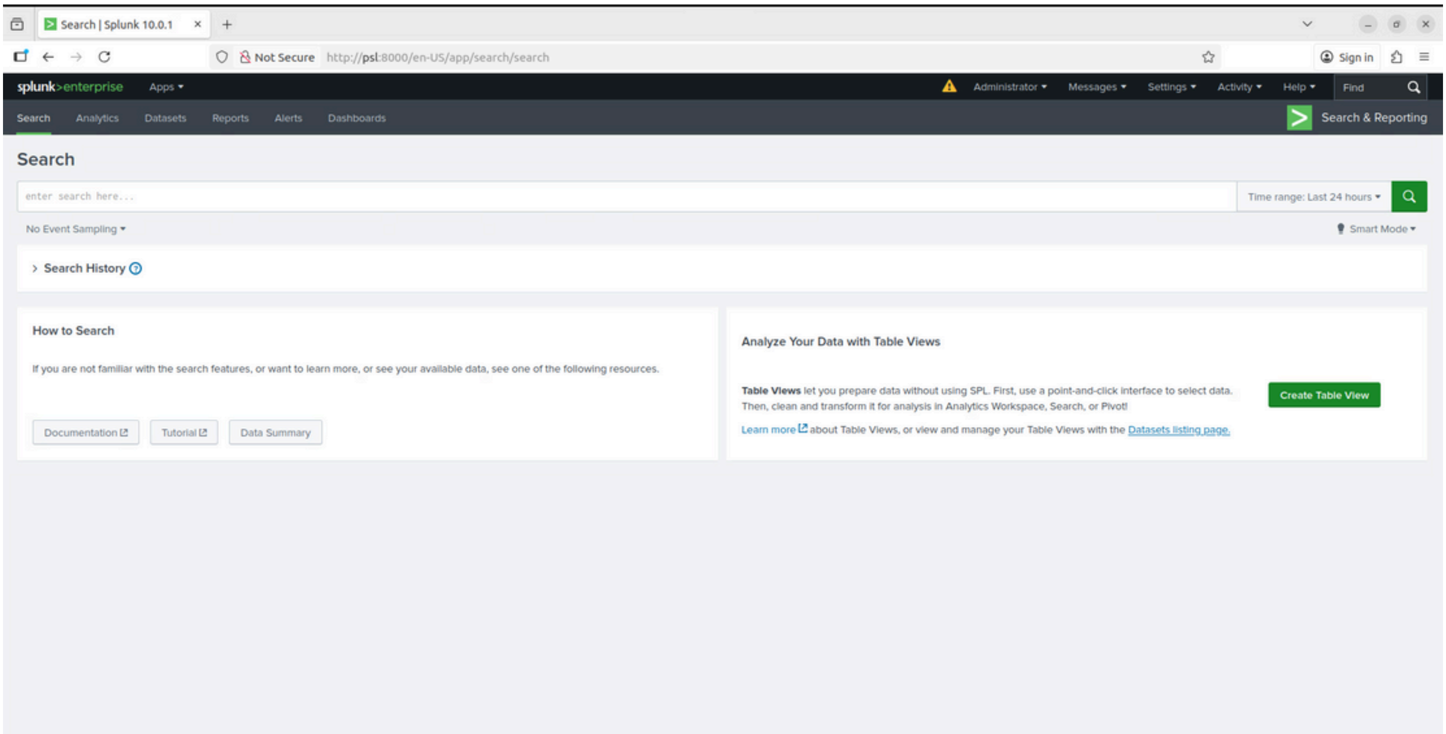
To disable Splunk from starting automatically at boot:

sudo systemctl disable splunk

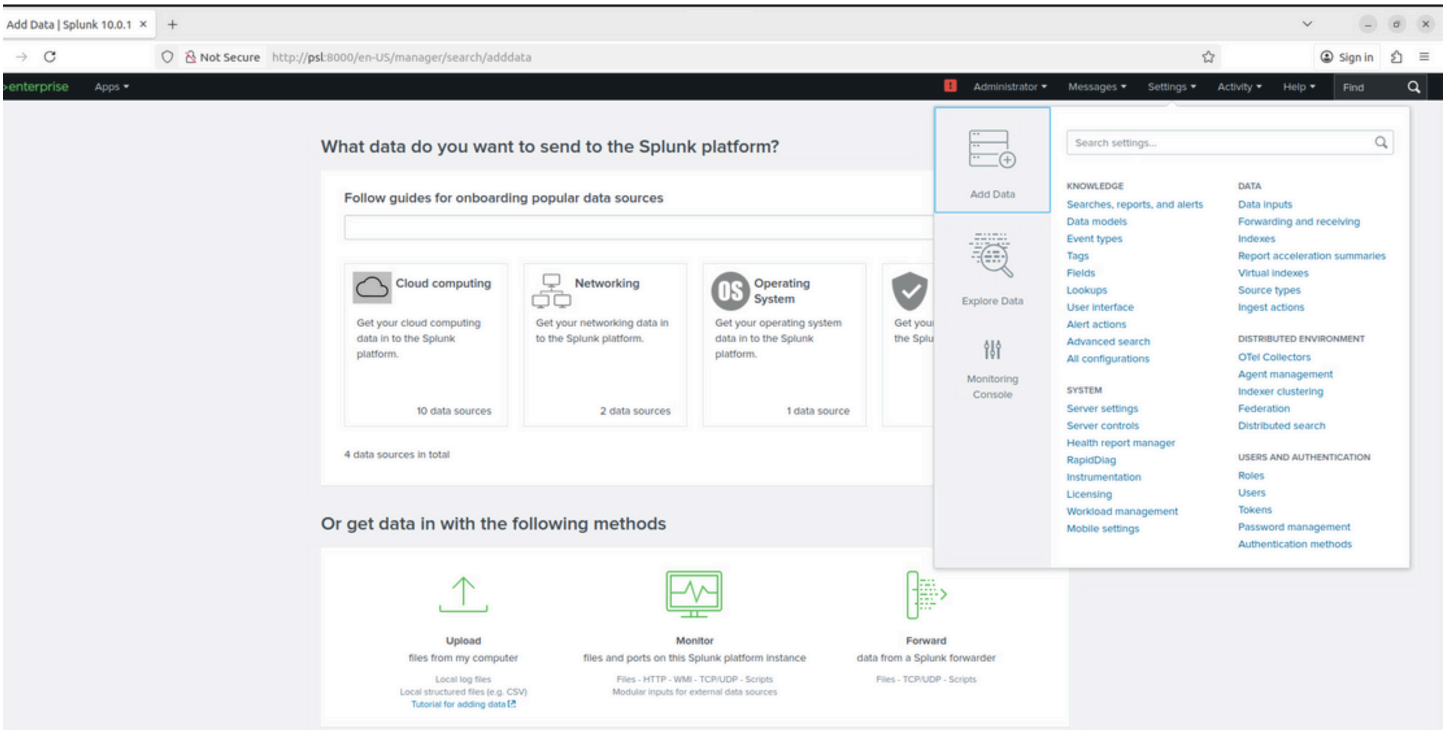
If needed, you can re-enable it later with:

sudo systemctl enable splunk





Now Import logs



Log File Dataset

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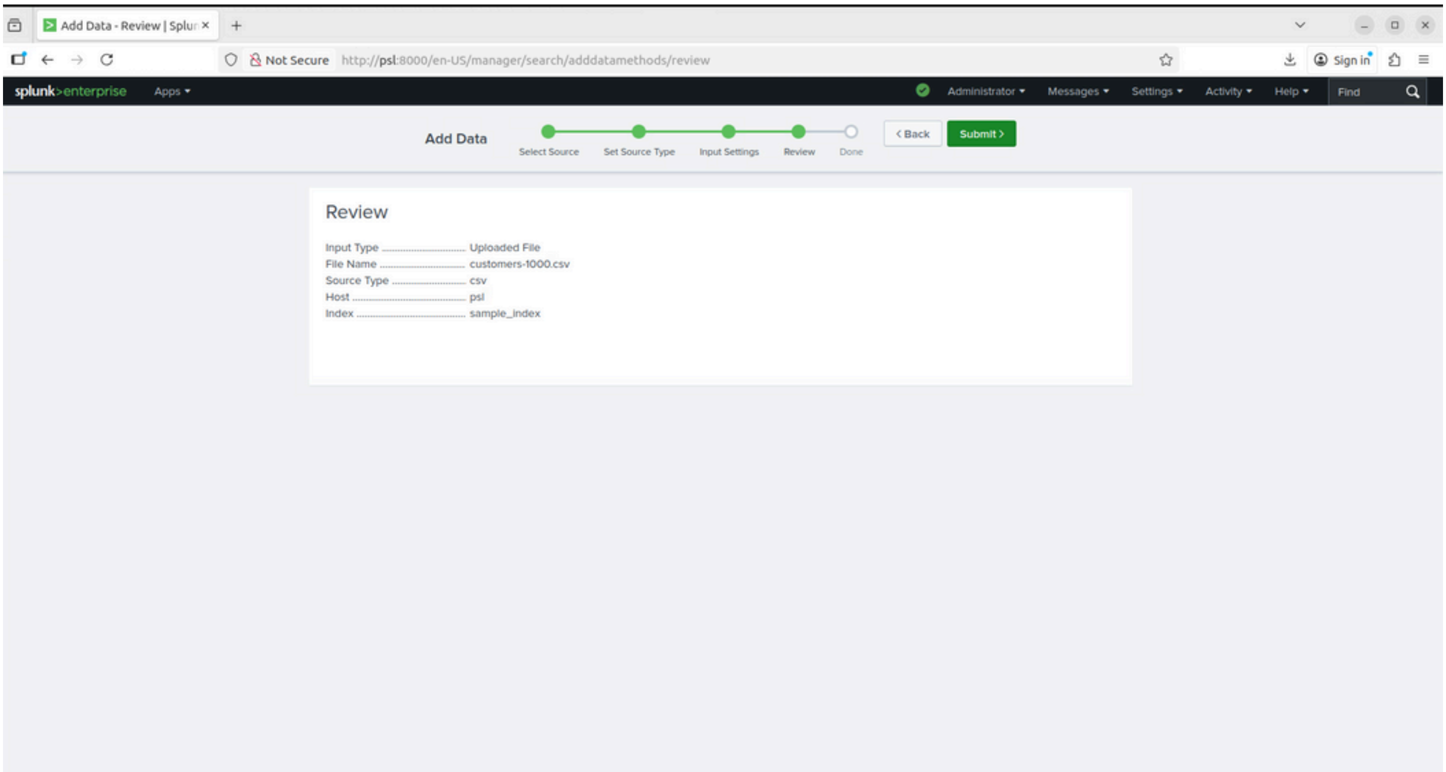
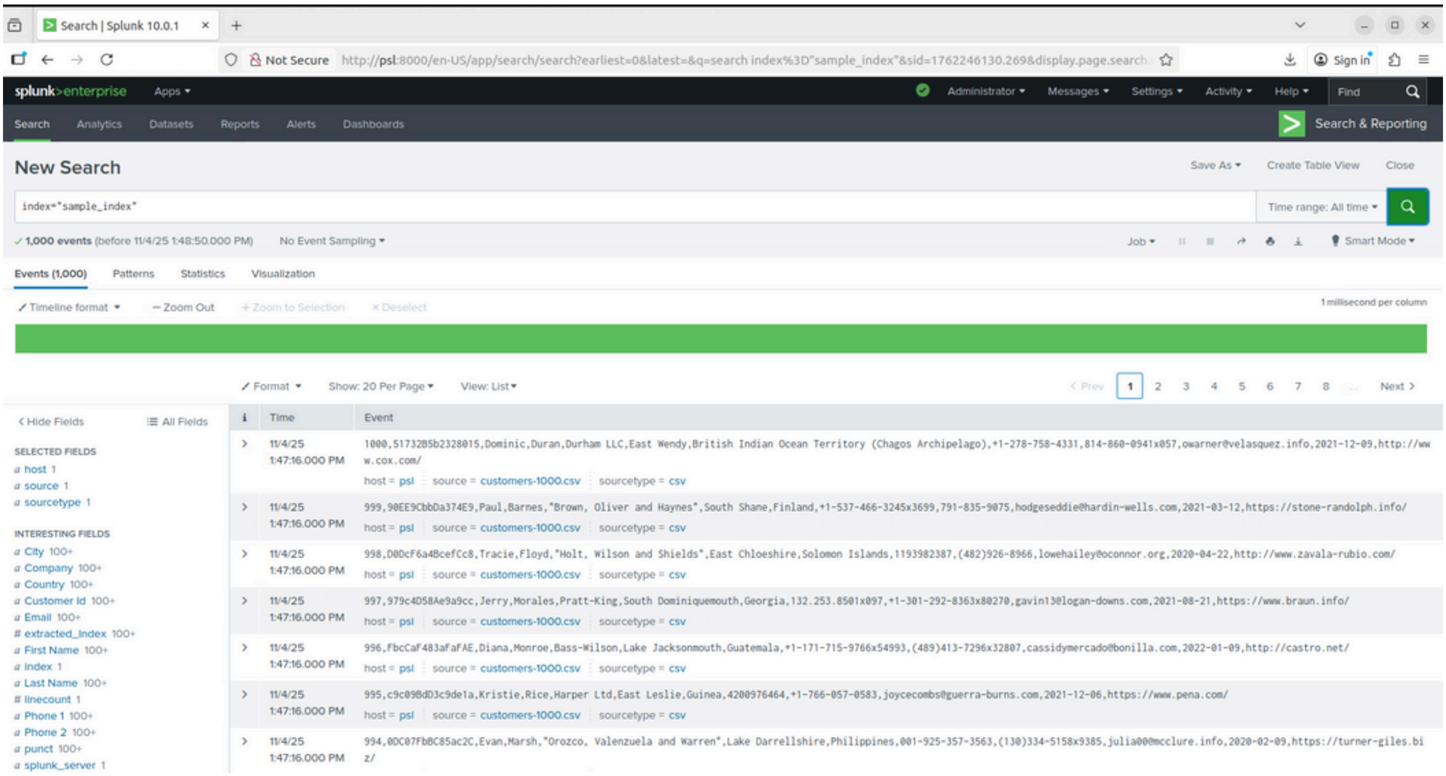


GO to search and report



Search Processing Language

When searching data in Splunk, there may be times when you have done your initial query using some search term, but you needed to filter some more to get some specific result set. How would you do that?

The answer is to use **SPL.**

What is SPL

SPL is Splunk's search language. It contains many commands, functions, arguments to help you get the desired result when searching a large dataset.

SPL has following components.

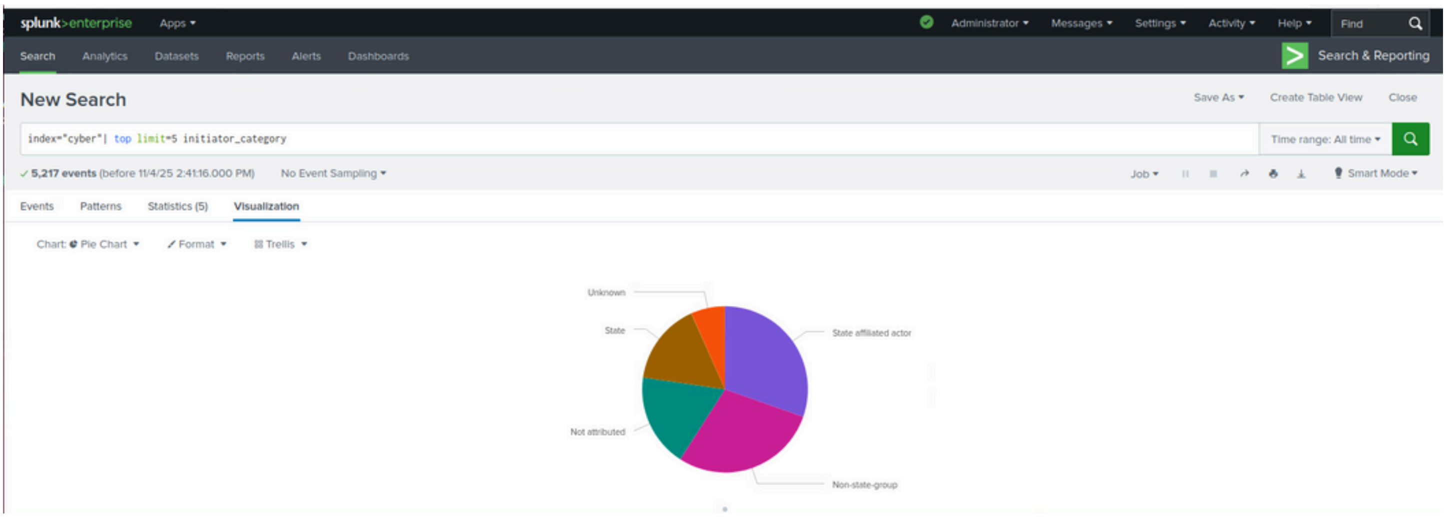
- **Search Terms** — these are the keywords/Phrases/Booleans that you mention in the search bar to get specific records from the dataset.
- **Commands** — once you get the result set, you want to do something with it, for example create a chart, compute stats, etc. You will write commands for this purpose.
- **Functions** — take input from a field being analysed and give the output after applying the calculations on that field.
- **Arguments** — functions are applied to the variables called Arguments in SPL, for example you might want to calculate the average for a specific field.
- **Clauses** — when you want to rename a field or group values by or over.

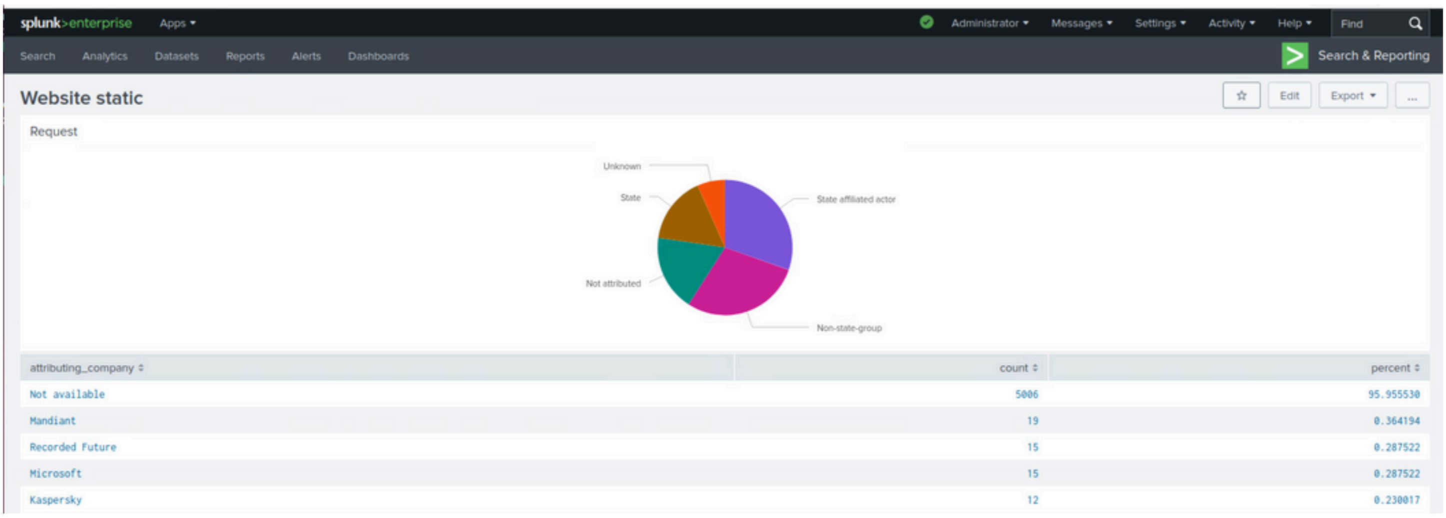
Splunk: Exploring SPL

Learn and explore the basics of the Search Processing Language.

tryhackme.com

CREATING DASHBOARD

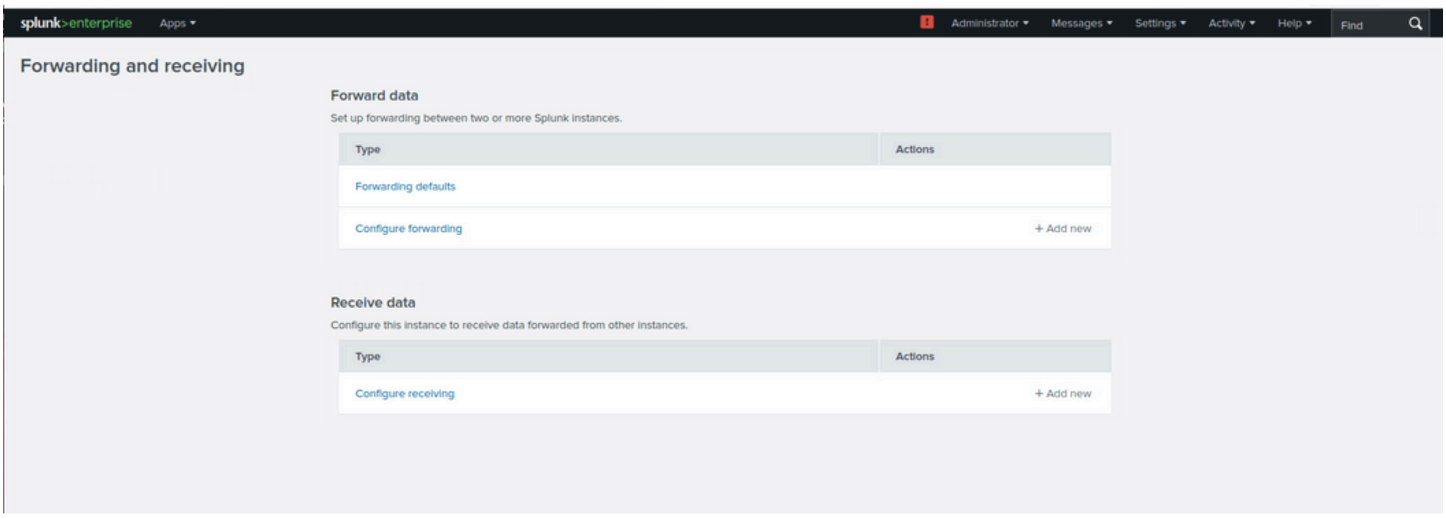




Challenge URL

1. https://bots.splunk.com/
2. https://bots.splunk.com/event/3oQ7sqI5bajOCP43o0svqT
   scenario/5vqgZJanGgTCXawi4nliIF

Deploy Forwarder



Configure receiving

Download receiver